

LENIN GOUD ATHIKAM

AI ENGINEER | MACHINE LEARNING | GENERATIVE AI & LLM SYSTEMS

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building Retrieval-Augmented Generation (RAG) systems, multi-agent LLM applications, and end-to-end machine learning pipelines. Skilled in Python, LangChain, PyTorch, TensorFlow, and cloud-based AI deployment, with research experience securing and evaluating LLM systems against adversarial attacks. Combines academic research with practical project delivery to design scalable, production-ready AI solutions.

TECHNICAL SKILLS

Programming: Python, SQL, R

AI Product Development & Prompt Engineering: RAG System Design, Prompt Engineering, Context Engineering, Structured Outputs (JSON Schema), Tool Calling, AI Agents, Multi-Agent Systems, MCP, A2A

Generative AI, NLP & LLM Frameworks: LangChain, LangGraph, CrewAI, OpenAI API, Hugging Face Transformers, NLP, Microsoft Foundry

Machine Learning & Deep Learning: Regression, Classification, Clustering, Recommendation Systems, Feature Engineering, Model Evaluation, Neural Networks, Predictive Modeling

Vector Databases & Semantic Retrieval: ChromaDB, FAISS, Vector Databases, Semantic Retrieval, PostgreSQL, MongoDB, SQLite

LLM Evaluation, Observability & MLOps: LangSmith, Ragas, Prompt Evaluation, LLM Tracing, Experiment Tracking, Model Deployment, Scalable AI Systems

Cloud AI Services & Deployment: AWS (EC2, Lambda, SageMaker), Hugging Face Spaces, REST APIs, Docker, CI/CD

Frameworks & Tools: FastAPI, Streamlit, Pandas, NumPy, Matplotlib, Power BI, Git/GitHub, TensorFlow, PyTorch

EXPERIENCE

Research Assistant

Jan 2026 – Apr 2026

Michigan Technological University | Houghton, MI

- Strengthened Retrieval-Augmented Generation (RAG) system security by researching Secure RAG architectures and designing defensive validation mechanisms, improving robustness against prompt injection and jailbreak attacks.
- Investigated adversarial attack vectors—including prompt injection, retrieval poisoning, and unauthorized information disclosure—to inform safer LLM deployment practices.

Computational Intelligence Teaching Assistant

Aug 2025 – Dec 2025

Michigan Technological University | Houghton, MI

- Guided 70+ students through machine learning, neural network, transformer, and large language model concepts, strengthening their applied AI development skills.
- Delivered technical mentorship on model implementation and evaluation methodologies, supporting coursework, grading, and consistent learning outcomes.

Data Analyst Intern

Aug 2023 – Sep 2023

EduLyt India (Airkrit India) | Remote

- Improved reporting accuracy and operational analytics by analyzing 100K+ student engagement records using SQL and Excel.
- Built interactive Power BI dashboards for KPI tracking, trend analysis, and stakeholder reporting, enabling faster data-driven decisions.
- Increased student retention strategy effectiveness by 12% through identification of key behavioral engagement patterns.

PROJECTS

AI Digital Twin Platform — Python, LangChain, ChromaDB, Hugging Face, RAG

[View Project](#)

- Delivered a production-ready conversational digital twin platform by integrating LLMs, contextual memory, and RAG pipelines with ChromaDB vector search and optimized chunking strategies.
- Deployed the application on Hugging Face Spaces with real-time inference and API integration, enabling scalable AI interaction workflows.

AI-Powered Agentic Workflow for Project Management — Python, CrewAI, LangChain, OpenAI API

[View Project](#)

- Delivered a multi-agent workflow automation system for project management by designing autonomous agent collaboration and task orchestration using CrewAI and LangChain.
- Built specialized AI agents for requirement analysis, task planning, progress tracking, and report generation, with memory-aware communication and tool calling to reduce manual coordination effort.

Multi-Agent Inventory & Quotation System — Python, SQLite, Agentic AI, Multi-Agent Systems

[View Project](#)

- Improved operational efficiency by designing a multi-agent sales and inventory management system for customer inquiry handling, quote generation, and order fulfillment.
- Built an orchestrator agent coordinating specialized agents for pricing, inventory validation, and supplier estimation, with SQLite-based tools enabling real-time fulfillment updates.

EDUCATION

Master of Science in Data Science

Apr 2026

Michigan Technological University | Houghton, MI

Bachelor of Technology in Computer Science (AI Specialization)

May 2024

Parul University | India